RL-KDA: A K-degree Anonymity Algorithm Based on Reinforcement Learning

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Abstract. K-degree anonymity is one of the privacy-preserving techniques and has gained attention in academia, industry, and government. Many social network data publishing algorithms based on K-anonymity techniques have been proposed, but most studies focus on static social networks. To address the existing problem of dynamic social networks, we propose a K-degree anonymity dynamic data publishing algorithm based on reinforcement learning. The algorithm ends with two phases: anonymization sequence and graph modification. In the anonymous sequence phase, this paper combines the idea of reinforcement learning and the characteristics of dynamic data change to build a reinforcement learning model for anonymous sequences. In this way, an ideal anonymous sequence can be created. We also propose a new strategy for graph modification, which selects edges according to degree centrality to generate anonymous graphs. Finally, experiments on real datasets show that this algorithm can reduce information loss and improve data utility.

Keywords: Privacy-preserving, K-Degree Anonymity, Reinforcement Learning, Social Network.

1 Introduction

Reinforcement learning is an unsupervised method for obtaining optimal strategies in a given environment [1]. Reinforcement learning is one of the solutions to the sequence generation problem. The generation problem of anonymous degree sequences is the combinatorial optimization problem, which in most cases is related to the decision order like the decision problem of sequences [2]. Since reinforcement learning has apparent advantages in sequence decisions, reinforcement learning can also solve the anonymous degree sequence generation problem. Considering how to strike a balance between protecting privacy and maintaining data utility is a crucial issue [3]. K-anonymity [4] is a strong notion of privacy that can be used to protect users' information. K-anonymity prevents attackers from identifying each user from (K-1) other users, identifying the user, and preventing attacks based on background knowledge [5]. But K-anonymity technique cannot be applied directly to graph data, so a more robust privacy model of

K-anonymity, K-degree anonymity [6], is gradually becoming one of the mainstream techniques for privacy protection in social networks.

To improve data utility, we introduce the idea of reinforcement learning. We propose a K-degree anonymity algorithm based on reinforcement learning to solve the problem of high information loss and low data utility. The main contributions of this paper are summarized as follows.

(1) RL-KDA uses the core idea of reinforcement learning and adopts the reward mechanism to find a proper sequence of anonymity eventually, reducing the degree of information loss and effectively improving data utility.

(2) A new graph modification strategy is proposed to use the degree centrality metrics of social networks for edge selection, which aims to minimize the changes to the original social networks in the anonymous graph generation process.

The rest of this paper is organized as follows. Section 2 introduces related work on graph data anonymization. Section 3 presents some related basics and the definition of K-degree anonymity graphs. Section 4 details the Markov modeling process of RL-KDA. Section 5 describes the detailed process of RL-KDA. Section 6 analyzes and summarizes the experimental results; finally, Section 7 is the conclusion of this paper.

2 Related Work

Although privacy-preserving models have been proposed successively, such as l-diversity [7], t-closeness [8], and differential privacy [9]. They cannot be directly applied to solve the privacy leakage problem because they are mainly used to solve the privacy protection problem of relational data publishing. The leading solution for social network graph data publishing is the K-degree anonymity technique [6], so we focus on the current state of research on graph modification-based K-degree anonymity technique in this part.

As one of the main methods of the K-degree anonymity technique, the graph modification technique changes the graph by modifying the nodes and edges to meet the privacy requirements. K-degree anonymity techniques based on graph modification are two categories. One type of research only modifies the edges and does not change nodes [10]. Liu et al. [6] first proposed the notion of K-degree anonymity, which means that a graph is K-degree anonymous if the degree of each node in the graph is at least K other nodes have the same degree. They proposed a dynamic K-degree anonymization two-step algorithm for Priority, which has the advantage of being applied to dynamic network datasets. However, it reduces the usefulness of the graph and does not apply to large networks because of the large information loss due to the anonymized degree sequence algorithm. In order to reduce the error caused by excessive modification of edges during graph anonymization, Lu et al. [11] proposed a fast-greedy algorithm (FKDA). The anonymous social network is made by simultaneously anonymizing the degree and adding edges, but the use of the greedy algorithm leads to the running time increases. Jordi et al. [12] proposed the UMGA algorithm, which uses univariate micro-aggregation to make anonymization graphs and selects edges by calculating the

neighborhood centrality score of each edge. It uses an exhaustive method in the process of the anonymity sequence, which leads to running time increases as K increases.

Another kind of research method is to modify nodes and edges simultaneously. For example, Chester et al. [13-14] proposed an algorithm to add virtual nodes, which achieves K-degree anonymity by adding new nodes and connecting them. Although the number of nodes and edges added is small, the original graph structure is still not better preserved because the addition of nodes makes the difference between the anonymized graph and the original graph larger. Ma et al. [15] proposed a KDVEM algorithm to achieve K-degree anonymization by modifying the graph structure in 2015. Firstly, we find the target degree for each node in the graph; secondly, some edges are added between the nodes to increase the degree of the nodes. If there exist vertices that are still not anonymous, use the Chester algorithm [13]. Thus, this algorithm can add the minimum number of new vertices and edges so that all vertices satisfy the anonymity requirement and preserve the graph structure to the maximum extent. However, the addition of virtual nodes distorts the graph structure, which makes the anonymous graph less utility.

To sum up, it can be found that existing K-degree anonymity algorithms are currently focused on the static social network, while less research is conducted on dynamic social network graph data. Therefore, graph modification-based K-degree anonymization technique is used for dynamic social network graph data in this paper. Compared with other anonymization techniques, the graph modification-based K-degree anonymization technique is simple and easy to implement. In addition, graph modification can retain more original graph features.

3 PRELIMINARIES

In this paper, we propose a new method for K-degree anonymity based on reinforcement learning, the purpose of which is to generate anonymity sequences with minimum of information loss. we consider the undirected graph $G = (V, E)$ with both edges and nodes without labels and weights, where V is the vertex, representing an individual in the social network, and E is the edge connected between two nodes, representing the connection between users. A more precise definition of K-anonymity and K-degree anonymity is as follows.

Definition 1. *(K-degree anonymity [6]). Assuming that each component $V_i \in V$ appears at least K times, then we say that the vector V satisfies K-degree anonymity.*

Definition 2. *(K-degree anonymity graph [6]). For an undirected graph $G = (V, E)$, if its degree sequence d satisfies K-degree anonymity, then the graph G is a K-degree anonymity graph.*

Definition 3. *(The anonymous degree sequence generation problem [6]). Given an undirected graph G and its degree sequence d, then an anonymous degree sequence $\tilde{d}$ is generated by K values, which in turn constitutes an anonymous graph $\tilde{G}$, then the distance between the graph G and the anonymous graph $\tilde{G}$ is defined by Equation (1).*

$$\mathrm{dis}(\mathrm{G}, \widetilde{\mathrm{G}}) = \sum_{\mathrm{i}=1}^{\mathrm{n}} \left|\mathrm{d}_{\mathrm{i}} - \widetilde{\mathrm{d}}_{\mathrm{i}}\right| \tag{1}$$

where the lower the value of dis*, the less information loss in the social graph and the higher data utility.*

4 Formal Modeling

4.1 Problem Description

Any reinforcement learning can' t be separated from Markov decision process. Markov Decision Processes (MDP) is a five-tuple $(A,\ P,\ S,\ R,\ \gamma)$, let the set A be the set of all actions in the environment, then $A = \{add,\ sub\}$, S denotes the set of states, $S = \{larger,\ smaller,\ no\ change\}$, s denote the current state and s' denotes the next state after taking action a; P denotes the transfer probability matrix, the set of probabilities of reaching state s' after taking action a in the current state s; let R denote the reward function, then $R = \{-1,\ +1,\ 0\}$; let γ denote the decay function, usually $\gamma \in (0,1]$.

The specific process is as follows: assume that the agent is in an environment E with a state space S. An arbitrary state $s \in S$ is a description of the agent's perceived environment. As in the decision process, the current state s is a known value, and the next state s' is an unknown value, and this value is a description of the error change in the state space. As in the anonymity sequence generation problem, it describes the changes of the difference in information loss in the current release. The action space A is composed of the actions taken by the Agent, such as increasing or decreasing the degree in the process of anonymity sequence generation. Suppose the action is taken in the current state, an action $a \in A$. The potential probability moving from the current state to the next state based on a probability, for example, in the state where the information loss degree becomes larger. In that case, if sub is chosen, the information loss difference of the graph will become smaller. However, the information loss difference may also not change in this paper because the information loss difference becomes larger, smaller and no change. It refers to the result of comparing the difference between the information loss at this moment and the information loss at the last moment obtained after increasing or decreasing the degree of the reinforcement learning process.

4.2 Problem solving model

This paper presents an optimal anonymity sequence policy using the Markov decision process. However, there is a problem with the establishment of the MDP model. The possibility of transition from the current state to the next state cannot be accurately determined, and thus the transfer probability in MDP cannot be determined. However, based on the study and research of reinforcement learning, using the time-series difference method can be an excellent solution to the difficulty of determining the transfer probability. The time-series difference method combines the Monte Carlo algorithm and dynamic programming, making it possible to solve the reinforcement learning

problem without using a complete sequence of states. Thus, it is overcoming the possibility that the state transitions cannot be determined in the model. Therefore, the state value function is computed using the following Equation (2).

$$V^*(s_t) = V^*(s_t) + \alpha[R_{t+1} + \gamma V^*(s') - V^*(s_t)] \quad (2)$$

where $V^*(s_t)$ is the value of the current state value function, $V^*(s')$ is the value of the state value function after a certain action is taken, and $\alpha \in (0,1]$ is the learning step. The action value function from Equation (3) as follows:

$$Q(s,a) = Q(s,a) + \alpha[R + \gamma(s',a') - v] \quad (3)$$

5 RL-KDA

In order to protect the privacy of users in undirected social networks, this paper proposes a reinforcement learning K-degree anonymity algorithm (Reinforcement Learning K-Degree Anonymity, RL-KDA), the algorithm is mainly divided into the following two steps.

We use the concept of K-degree anonymity to protect user privacy and modify only the edge set to create a new anonymity graph $\tilde{G} = (V, d)$, where $\tilde{G}$ represents the anonymity graph, as implemented in Algorithm 1.

Algorithm 1: RL-KDA

Input: The original graph G, Anonymous parameter K
Output: The anonymous graph $\tilde{G}$
1: d= the degree sequence of the original graph G
2: anonymous sequence $\tilde{d}$ = RL_DegreeSequence(d)
3: anonymous $\tilde{G}$=Graph_modification($\tilde{d}$,G)
4: **return** The anonymous graph $\tilde{G}$

Algorithm 1 describes the overall process of RL-KDA, where the input is the original graph and the privacy parameter K, and the output is the anonymized graph. Line 1 describes the original degree sequence from the original graph; line 2 represents the process described in Section 5.1 to generate the anonymous degree sequence using reinforcement learning. Line 3 describes the new graph modification algorithm, which is implemented as in Section 5.2. Moreover, line 4 returns the anonymous graph.

5.1 Anonymity Degree Sequence

The first part of the work of RL-KDA is to anonymize degree sequences based on degree sequences of graphs. In this paper, we introduce the idea of reinforcement learning to generate anonymous degree sequences. The algorithm for this part is described in detail below.

Algorithm 3: RL_Degree sequence

Input: Original degree sequence $d, Q(s,a), \forall s \in S$, $\forall a \in A$, $R = \{-1, +1, 0\}$,α,γ,$t_i \epsilon T$
Output: Anonymity sequence at time t_{i+1}

1: $\tilde{d}_{t_i}$=greedy_partition(d) // Anonymity sequence at time t_i
2: $dis_1 = |sum(\tilde{d}_{t_i}) - sum(d)|$
3: Initialize $Q(s,a) = 0, s = s_0$
4: **if** $\tilde{d}_{t_i}$ does not satisfy the privacy level requirement at t_{i+1} **then**
5: Randomly select a at t_i and get s and R
6: **for** i=1,2…, **do**
7: **if** $sum(\tilde{d}_{t_i}) > sum(d)$ **then**
8: choose add from A
9: **else**
10: choose sub from A
11: **end if**
12: $dis_2 = |\mathrm{sum}(\tilde{\mathrm{d}}_{\mathrm{t_{i+1}}}) - \mathrm{sum(d)}|$
13: $s \leftarrow s'$
14: compare dis_1 with dis_2
15: **if** s="$larger$" **then**
16: R=-1
17: **else if** s="$smaller$" **then**
18: R=+1
19: **else**
20: R=0
21: **end if**
22: **end for**
23: **end if**
24: $\mathrm{q(s,a)} \leftarrow \mathrm{q(s,a)} + \alpha + (R + \gamma + \mathrm{q}(s', a') - \mathrm{q(s,a)})$
25: add q(s, a)to Q(s, a)
26: until q(s, a) is convergence
27: choose maximum q(s, a) ∈Q(s, a)
28: $\pi = \underset{a}{\mathrm{argmax}}\ q(\mathrm{s,a})$

Finding an ideal anonymity sequence assignment scheme through continuous iteration in the anonymity sequence generation. In the iteration process, this paper uses the characteristics of dynamic data combined with the Markov decision process to select the corresponding action. Suppose the anonymity sequence of t_i does not meet the privacy level requirement of t_{i+1}. In that case, a new anonymity sequence should be generated to meet the privacy protection requirement, so the action of $\{add, sub\}$ can be chosen. In addition, we perform a series of operations based on the anonymity sequence and obtain the optimal anonymity sequence by continuously updating the value function. Next, the pseudo-code of the specific algorithm is given.

In Algorithm 2, the sequence of anonymity degrees produced at the moment using the greedy algorithm [15] in line 1. dis_1 in line 2 represents the degree of information loss between the anonymous degree sequence and the original degree sequence at the moment t_i, and line 3 initializes the parameters. If the anonymity degree sequence at t_i does not satisfy the privacy requirement at t_{i+1} in lines 4-5. First, the method selects action randomly and obtains the corresponding state and reward at the first time point. In lines 6-11, the sum of the anonymous degree sequence is compared with the sum of

the initial degree sequence in two cases: the first one is that if the sum of the anonymous degree sequence is less than the sum of the initial degree sequence, action add is taken. If the sum of the anonymous degree sequence is greater than the sum of the initial degree sequence, take sub action. Lines 12-23, based on the previous anonymous degree sequence, compare the two sums of degree sequences at the release time point starting from t_{i+1}. The state is updated according to the comparison result. If the state after taking action changes to $larger$, then $R = -1$. If the state after taking action changes to $smaller$, $R = +1$. If the same degree of information loss as the original graph, i.e., the state after taking action becomes $no\ change$, then $R = 0$. In lines 24-26, the value function is updated by the initial value of the value function and the reward value after selecting the action, storing each updated value, and keeping iterating until the value function converges. In lines 27-28, we choose the largest value in the value function and the strategy to obtain the maximum value function. The degree sequence minimizes information loss at this point.

5.2 Graph Modification

In the graph modification, we can select auxiliary edges based on the relevance of each edge. By doing this, we can remove or add edges that create a low correlation, thus reducing information loss in the process. If edges with a low correlation between edges are deleted, then the important edges in the resulting anonymity network are retained. Therefore, it is crucial to decide on a relevant strategy to reduce the information loss. In this paper, we use Degree Centrality to select edges by first choosing the nodes that need to be modified. Then, using Degree Centrality to sort these nodes, the smaller the Degree Centrality is, the higher the ranking is. This strategy maximizes the availability of dynamically published data by retaining as many important edges as possible.

Degree centrality is the most direct measure of node centrality in network analysis. The greater the degree of the node means the higher the degree centrality of the node, and the more influential the node is in the network. The degree centrality value of a node is calculated as follows.

$$DegreeCentrality(v_i) = \frac{\deg(v_i)}{count(V) - 1}\ (v_i \epsilon V) \tag{4}$$

where $\deg(v_i)$ represents the degree of node v_i and $count(V)$ represents the number of nodes in the graph.

In the graph modification, we select the nodes that need to be modified. Then, the nodes are sorted by degree centrality, and the nodes with small degree centrality are sorted first, so preserve essential edges as much as possible to maximize the utility of dynamically published data. Finally, RL-KDA adds edges, switches edges, and deletes edges [12]. The pseudo-code of the graph modification algorithm is given below.

In Algorithm 3, line 1 calculates which node degrees need to be changed, and line 2 sorts the nodes in the node list $changeNodes$ that changed in ascending order according to the degree center. Lines 3-7, when the element in the list is less than 0, it means the node needs to add an edge, add the node to the list of nodes that need to add an edge $nodesAdd$; conversely, it means the node needs to delete an edge, add the node to the list of nodes that need to delete an edge $nodesDel$. In lines 8-14, compared the length

of $nodesDel$ with $nodesAdd$. If the number of nodes to add edges is greater than the number of nodes to delete edges, then add edges. If the number of nodes to add edges is less than the number of nodes to delete edges, then the delete edge operation is performed. If the number of nodes to add edges is equal to the number of nodes to delete edges and not equal to 0, then the switch edge operation is performed. Until the length of the list of nodes that need to add edges and the length of the list of nodes that need to delete edges are equal to 0, return the anonymous graph.

Algorithm 3: Graph_modification

Input: The anonymity sequence $\tilde{d}$, K
Output: The anonymity graph $\tilde{G}$
1: $\tilde{G}$=G.copy()
2: changeNodes=$\tilde{d}[i] - d[i]$,premution= []
3: changeNodes order by degree Centrality, premution.append(v_i)
4: **for** i in range(len(changeNodes))
5: **if** changeNodes[i]<0 **then**
6: nodesDel.append(premution[i])
7: **else**
8: nodesAdd.append(premution[i])
9: **end if**
10: **end for**
11: **if** len(nodesDel)<len(nodesAdd) **then**
12: $\tilde{G}$.Addedge()
13: **else if** len(nodesDel)=len(nodesAdd)≠0 **then**
14: $\tilde{G}$.SwitchEdge()
15: **else**
16: $\tilde{G}$.DeleteEge()
17: **end if**
18: **return** The anonymity graph $\tilde{G}$

6 Experimental results

6.1 Datasets

Table 1 Experimental datasets.

Datasets	Edges	Nodes	ACC	T	APL
Polbooks	441	105	0.487	0.348	0.308
Ca-AstroPh	198110	18772	0.677	0.318	4.194
Ca-CondMat	93497	23133	0.633	0.264	5.352

In this paper, Table 1 shows the properties of these three datasets in experiments. The first dataset is Polbooks, a network of American political books published; these last two datasets are from Arxiv. All of the above datasets can be available on Stanford University's website [16].

6.2 Evaluation Metrics

To evaluate the effectiveness of RL-KDA, we use the metrics to measure the amount of distortion of the anonymized graph in this section. These metrics are some features of the graph, such as T (Transitivity), ACC (Average Clustering Coefficient), and APL (Average Path Length), if the values of these three features are closer to the feature values of the original graph. Then it means data utility of the graph higher [18].

Transitivity: It is actually to calculate the ratio of the triangles in the whole graph and the three connected triangles, which reflects the ratio of the triangles in the graph, and is defined as follows.

$$\mathrm{T} = \frac{3 * \Delta}{\Lambda} \tag{5}$$

where Δ is the number of triangles and Λ is the number of connected triangles.

Average Clustering Coefficient: The clustering coefficient of a vertex is the proportion of possible triangles that exist through that vertex, and the average clustering coefficient of a graph is actually the average clustering coefficient of all vertices. The definition is as follows.

$$\mathrm{ACC(G)} = \sum_{\mathrm{u}=1}^{\mathrm{n}} \frac{2\mathrm{T(u)}}{\mathrm{deg(u)} * (\mathrm{deg(u)} - 1)} \tag{6}$$

where $T(u)$ is the number of triangles passing through node u, $deg(u)$ is the degree of u.

Average Path Length: APL is the last metric of a graph, which measures the average of the shortest paths between nodes in the graph. The social distance between all connectable pairs of a graph is measured by the average path length (APL). It is defined as follows.

$$\mathrm{APL(G)} = \frac{\sum_{(\mathrm{u,v})\epsilon \mathrm{Pair}} \mathrm{SPL(u,v)}}{|\mathrm{P}|} \tag{7}$$

where P denotes all reachable vertex pairs and $SPL(u, v)$ denotes the shortest path length between vertices u and v. It is a measure of the efficiency of information or mass transport on the network.

6.3 Experimental analysis

In this paper, we tested comparison experiments in different datasets, where the algorithms compared are FKDA [11], UMGA [12], Vertexadd [13], KDVEM [15], TSRAM [18] and GA-KDA [19]. The fore four algorithms are more representative algorithms in K-degree anonymous social graphs. Since these four methods have been introduced in detail in Section 2 on related work, thus, we do not describe them here.

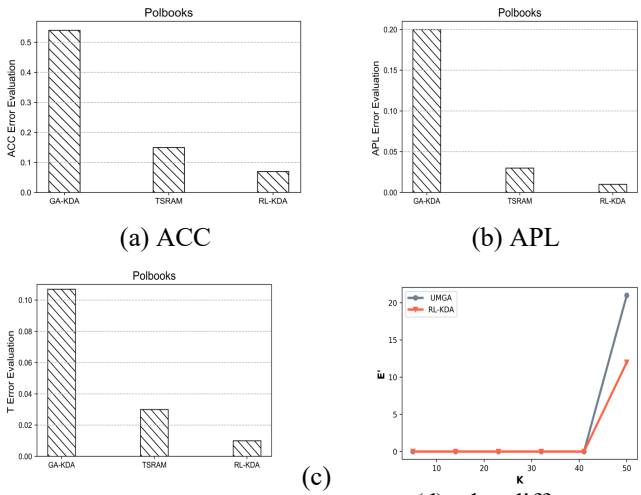
TSRAM generates a sequence of anonymity degrees of a graph based on a tree structure and uses bottom-up node partitioning of the graph based on anonymization levels to achieve dynamic changes in privacy protection levels. GA-KDA combines community detection and genetic algorithms to generate anonymous graphs. In the anonymity sequence generation algorithm, the learning rate $\alpha = 0.1$ and the decay parameter $\gamma = 0.09$. To ensure the experiment's accuracy, the results of the experiment are the mean of the results of 10 experiments.

In Figure 1(a) (b) (c), the horizontal coordinates are the GA-KDA, TSRAM，RL-KDA in that order, and the vertical coordinates are the error evaluation of each metric. The error evaluation of each metric in Figure 1 is calculated as [17].

$$EV = \frac{1}{n}\sum_{i=1}^{n}|a_i - a| \tag{8}$$

where n is the number of K values, which is represented as 13 in this section. a_i is the metric of the ith K-degree anonymized graph, and a represents the metric value of the original graph.

From Figure 1(a) (b) (c), it can be seen that RL-KDA performs better in comparison with both TSRAM and GA-KDA methods. For example, in Figure 1(b), GA-KDA achieves a transferability error of 0.1 or more, while TSRAM has an error between 0.02 and 0.04, while RL-KDA keeps the error below 0.02. In Figure 1(d), the horizontal coordinate K represents the value of the privacy parameter, and the vertical coordinate E' represents the difference between the edges of the original graph and the edges of the anonymized graph. E' is calculated as $E' = |E - \tilde{E}|$, where E and $\tilde{E}$ represent the edges of the original and anonymized graphs, respectively. Figure 1(d) mainly compares the edge errors of UMGA [24] and RL-KDA, and we can find that neither algorithm needs to add extra edges at K ∈ [5, 25]. However, it changes at K = 50, and the edge difference of UMGA is twice as that of RL-KDA, which indicates that RL-KDA preserves as much as possible the original graph structure at larger values of K.



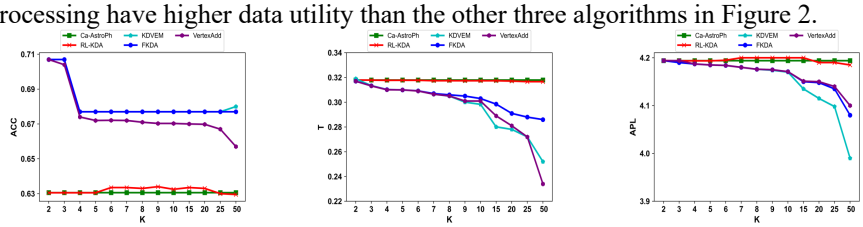
(a) ACC (b) APL

(c) T (d) edge difference

Fig. 1. Polbooks Error Evaluation.

Figure 2 shows the effect of K values on each evaluated metric on Ca-AstroPh. Figure 2(a) shows that the trend of the average clustering coefficients of the three algorithms like FKDA becomes smaller. However, because these three methods do not make reasonable improvements to the generation of anonymity sequences, it leads to the fact that fundamentally the average clustering coefficients of these three methods are much higher than the average clustering coefficient values of the original graphs. shows the effect of RL-KDA and the other three methods' transitivity of K values. From

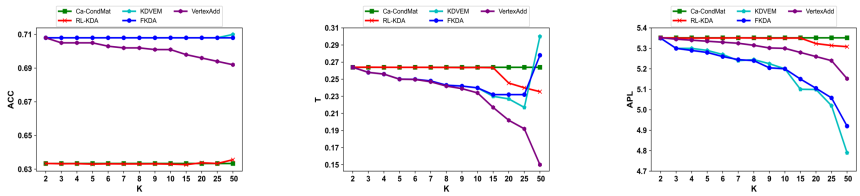
Figure 2(b), it can be seen that the values of RL-KDA and the transitivity of the original graph are basically the same. In Figure 2(c), RL-KDA is closest to the APL values of the original graphs. This indicates that the anonymity graphs obtained by RL-KDA processing have higher data utility than the other three algorithms in Figure 2.



(a) ACC (b) T (c) APL

Fig. 2. Ca-AstroPh.

Figure 3 evaluates the influence of K values on each metric on Ca-CondMat. In Figure 3(a), the ACC of RL-KDA and the original graph are almost the same, mainly because RL-KDA does not add new edges. From Figure 3(b), it can be seen that the value of transitivity tends to decrease when $K \geq 15$, which is due to the fluctuation of the value of transitivity due to the effect of the edge switch. Figure 3(c) compares the effects between the average path length (APL) metric and the K values before and after graph anonymization. a closer look reveals that RL-KDA is a slight decrease at K=15. This is because RL-KDA is a random modification of the graph, which leads to the removal of some important edges.



(a) ACC (b) T (c) APL

Fig. 3. Ca-CondMat

In a word, it can be found that RL-KDA is closer to the metrics of the original graph than other algorithms through comparison experiments. The reason why RL-KDA has advantages over the other four algorithms is that it reduces the information loss in the process of anonymity sequence, solves the problem of low data utility from the essence of the problem, and combines the new graph modification algorithm to improve data utility.

7 Conclusion

We propose a reinforcement learning-based privacy-preserving algorithm named RL-KDA, which combines reinforcement learning with the K-degree anonymization technique. RL-KDA can effectively improve the data utility of dynamic social networks while guaranteeing privacy protection for graph data. However, all social network

datasets in this paper are undirected graphs, while directed social networks are also a new research hotspot. Therefore, applying RL-KDA to directed social networks is also one of the leading works in the future.

Acknowledgment

This paper is supported by the Science and Technology Program of Inner Mongolia Autonomous Region (Grant No. 2019GG116).

References

1. Botvinick M, Ritter S, Wang J X, et al. Reinforcement learning, fast and slow[J]. Trends in cognitive sciences, 2019, 23(5): 408-422.
2. Ni Z, Paul S, Zhong X, et al. A reinforcement learning approach for sequential decision-making process of attacks in smart grid[C]//2017 IEEE Symposium Series on Computational Intelligence (SSCI). IEEE, 2017: 1-8.
3. Yin C, Xi J, Sun R, et al. Location privacy protection based on differential privacy strategy for big data in industrial internet of things[J]. IEEE Transactions on Industrial Informatics, 2017, 14(8): 3628-3636.
4. Sweeney L. k-anonymity: A model for protecting privacy[J]. International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems, 2002, 10(05): 557-570.
5. Hao Y, Cao H, Hu C, et al. K-anonymity for social networks containing rich structural and textual information[J]. Social Network Analysis and Mining, 2014, 4(1): 223.
6. Liu K, Terzi E. Towards identity anonymization on graphs[C]//Proceedings of the 2008 ACM SIGMOD international conference on Management of data. 2008: 93-106.
7. Machanavajjhala A, Kifer D, Gehrke J, et al. L-diversity: Privacy beyond kanonymity[J]. ACM Transactions on Knowledge Discovery from Data, 2006, 1(1):3.
8. Li N, Li T, Venkatasubramanian S. t-closeness: Privacy beyond k-anonymity and l-diversity[C]//2007 IEEE 23rd international conference on data engineering. IEEE, 2007: 106-115.
9. Dwork C. Differential privacy[C]//International Colloquium on Automata, Languages, and Programming. Springer, Berlin, Heidelberg, 2006: 1-12.
10. Zhou B, Pei J, Luk W S. A brief survey on anonymization techniques for privacy preserving publishing of social network data[J]. ACM Sigkdd Explorations Newsletter, 2008, 10(2): 12-22.
11. Lu X, Song Y, Bressan S. Fast identity anonymization on graphs[C]//International Conference on Database and Expert Systems Applications. Springer, Berlin, Heidelberg, 2012: 281-295.
12. Casas-Roma J, Herrera-Joancomartí J, Torra V. k-Degree anonymity and edge selection: improving data utility in large networks[J]. Knowledge and Information Systems, 2017, 50(2): 447-474.
13. Chester S, Kapron B M, Ramesh G, et al. k-Anonymization of Social Networks by Vertex Addition[J]. ADBIS (2), 2011, 789: 107-116.
14. Chester S, Kapron B M, Ramesh G, et al. Why Waldo befriended the dummy? k-Anonymization of social networks with pseudo-nodes[J]. Social Network Analysis and Mining, 2013, 3(3): 381-399.

15. Ma T, Zhang Y, Cao J, et al. KDVEM: A k-degree anonymity with vertex and edge modification algorithm[J]. Computing, 2015, 97(12): 1165-1184.
16. Stanford large network dataset collection. http://snap.stanford.edu/data/.
17. Rajabzadeh S, Shahsafi P, Khoramnejadi M. A graph modification approach for k-anonymity in social networks using the genetic algorithm[J]. Social Network Analysis and Mining, 2020, 10(1): 1-17.
18. Kiabod M, Dehkordi M N, Barekatain B. TSRAM: A time-saving k-degree anonymization method in social network[J]. Expert Systems with Applications, 2019, 125: 378-396.